

CSE 331

AUTOMATA AND COMPUTABILITY

Lecture 3

Summer 2026

Today's learning goals

Sipser Ch 1.1, 1.2

- Review what it means for a **set** to be closed under an **operation**.
- Define the regular operations on languages
- Prove closure properties of the class of regular languages

$$L = \{ w \mid \text{DFA accepts } w \}$$

$$w \in L : \text{DFA} \rightarrow \text{Yes}$$

$$w \notin L : \text{DFA} \rightarrow \text{No}$$

Building DFA

$\Sigma = \{0, 1\}$
3 bit memory
 $2^3 = 8$ states

000 = 0
001 = 1
010 = 2
011 = 3
100 = 4
101 = 5
110 = 6
111 = 7



Remember

↘ No real Memory like RAM
States = Memory
0(1)

$L = \{w \mid w \text{ when interpreted as a binary number is divisible by } 3\}$

States are our only (computer) memory.

We needed to keep track of 3 remainders.

Design and pick states with specific roles / tasks in mind.

Remainder = 0 [accept state]

"Have not seen any of desired pattern yet"

"Trap state"

$d=8$
 $0 \leq r < 8$
0, 1, 2, 3, 4, 5, 6, 7

$0 \leq r < d$ $d=3$
 $0 \leq r < 3$
0, 1, 2

$M = \{w : \text{after reading } w \text{ reaches final state}\}$

A useful (optional) bit of terminology

When is a string accepted by a DFA?

$M = \{w : \delta^*(q, w) \in F\}$

$$\delta(q, c) = q_{\text{next}} \\ q \in Q, c \in \Sigma$$

Computation of M on w : where do we land when start at q_0 and read each symbol of w one-at-a time?

$$\delta^*(q, w) = \begin{cases} q & w = \epsilon \\ \delta(q, w) & \text{if } w = c, c \in \Sigma \\ \delta^*(\delta(q, c), w') & \text{if } w = cw', c \in \Sigma, w' \in \Sigma^* \end{cases}$$

↙ δ^* processes the input symbol one at a time until no symbol left.

Recursively
defined
function

Prove that the given lang. L is regular.

Sipser p. 35 Def 1.5

Regular languages

- For DFA M over the alphabet Σ
 - For each string w over Σ , M either accepts w or rejects w
 - The **language recognized by M** is the set of strings M accepts
 - The **language of M** is the set of strings M accepts
 - **$L(M)$** = $\{ w \mid w \text{ is a string over } \Sigma \text{ and } M \text{ accepts } w \}$

$L(M)$

$w \in L : M \text{ accept}$
 $w \notin L : M \text{ rejects}$

Definition: A language is **regular** iff there is some finite automaton that recognizes **exactly** it.

$$L = \{ w \mid \text{DFA accepts } w \}$$

Regular languages: bounds?

$$L \subseteq \Sigma^*$$

Is **every** finite language regular?

$$L = \{w \mid \text{the length of } w \leq 2\}$$

$$\Sigma = \{0, 1\}$$

$$L = \{00, 01, 10, 11\} = \text{finite language}$$

A. No: some finite languages are regular, and some are not.

L is regular, because we can build a DFA.

B. No: there are no finite regular languages.

C. Yes: every finite language is regular.

D. I don't know.

*Every finite lang is regular,
but not every regular lang. is finite.*

$$L \subseteq \Sigma^*$$

Building DFA

New strategy

Substring, Prefix, Suffix

V is a Substring of $W \iff$ there exists strings x, y such that $W = xvy$
if $x = \epsilon$, then v is prefix of W .
if $y = \epsilon$, then v is suffix of W .



Express L in terms of simpler languages – use them as building blocks.

$$\bar{L} = \Sigma^* - L$$

Everything outside the lang. but still inside Σ^*

Example

$L = \{ w \mid w \text{ does not contain the substring baba} \}$

= the complement of the set

$\{ w \mid w \text{ contains the substring baba} \}$

$$\bar{L} = L^c = \{ w \mid w \text{ contain the substring baba} \}$$

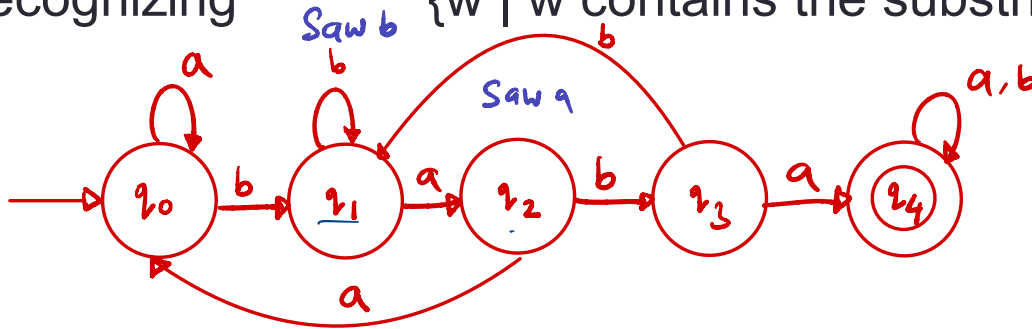
Building DFA

$\Sigma = \{a, b\}$



DFA recognizing

$\{w \mid w \text{ contains the substring } baba\}$



XXXX

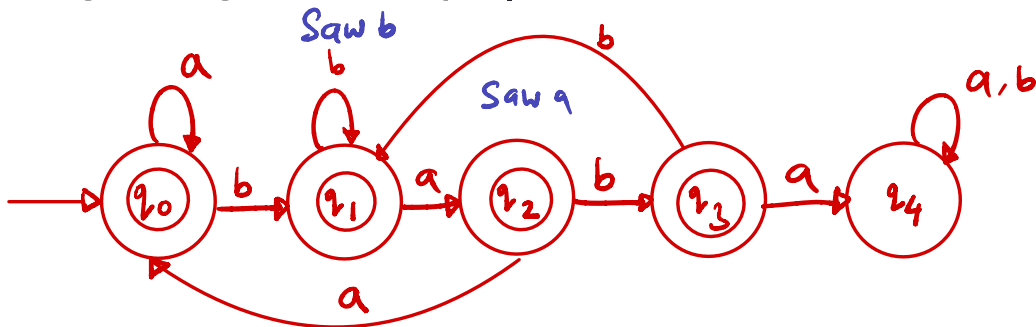
b b b b a b a

b a ~~aaa~~ b a

b a b ~~bbb~~ a

DFA recognizing

$\{w \mid w \text{ doesn't contain the substring } baba\}$



Swap final and reject states

Complementation

Theorem: If A is a regular language over $\{0,1\}^*$, then so is its complement $\overline{A}$

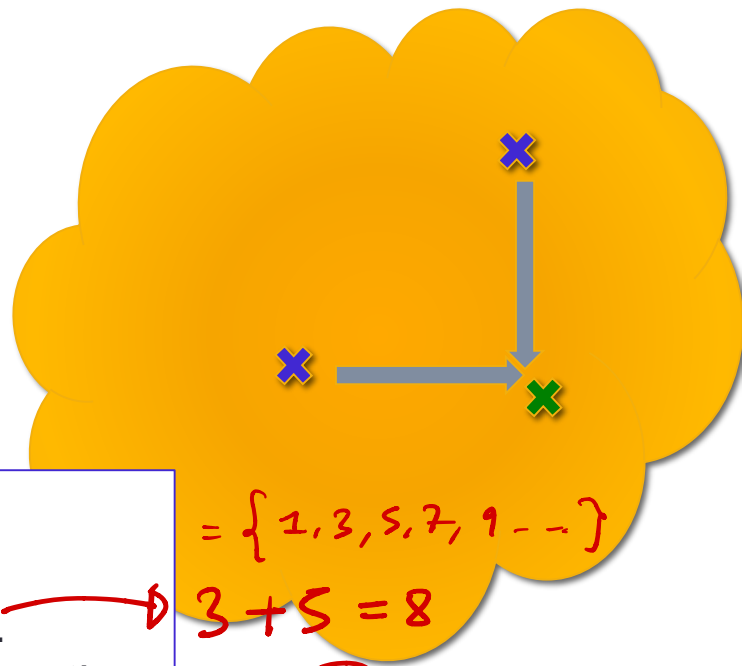
aka "the class of regular languages is closed under complementation"

Closure of ... under ...

- $\mathbb{Z}$ under addition.
 Set of integers $\mathbb{Z} = \{0, 1, 2, 3, 4, \dots\}$
 $1 + 2 = 3$
- Set of even ints under multiplication.
- ✓ $\{0\}^*$ under concatenation.

Which of these is true?

- ~~A~~ The set of odd integers is closed under addition.
- ~~B~~ The set of positive integers is closed under subtraction.
- ✓ C The set of rational numbers is closed under multiplication.
- ~~D~~ The set of real numbers is closed under division.
- ~~E~~ I don't know.



$$3 - 5 = (-2)$$

$$\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$$

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

$b, d \neq 0$

Complementation

Theorem: If A is a regular language over $\{0,1\}^*$, then so is its complement $\overline{A}$

aka "the class of regular languages is closed under complementation"

Complementation

Theorem: If A is a regular language over $\{0,1\}^*$, then so is its complement $\bar{A}$

aka "the class of regular languages is closed under complementation"

Proof: Let A be a regular language.

Then there is a DFA $M = (Q, \Sigma, \delta, q_0, F)$ such that $L(M) = A$.

We want to build a DFA M' whose language is $\bar{A}$. Define

$M' =$

?

Claim of Correctness $L(M') = \bar{A}$

Proof of claim...

A is regular, accepted by DFA $M = (Q, \Sigma, \delta, q_0, F)$

Define a new DFA $M' = (Q, \Sigma, \delta, q_0, Q \setminus F)$ ← Construct.

↘ $Q - F$ (Set of states without F, this is known as difference between sets)

Claim: M' accepts A^c

Proof: because M accepts A .

$$A = \{w : M \text{ accepts } w\} = \{w : \delta^*(q_0, w) \in F\}$$

$$A^c = \{w : w \notin A\} = \{w : \delta^*(q_0, w) \notin F\} = \{w : \delta^*(q_0, w) \in Q \setminus F\}$$

So M' accepts A' .

Why closure proofs?

- General technique of proving a new language is regular
- Stretch the power of the model
- Puzzle!

The regular operations

Sipser Def 1.23 p. 44

① Complement

For A, B languages over same alphabet, define:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\} \text{ Today}$$

$$A \circ B = \{xy \mid x \in A \text{ and } y \in B\}$$

$$A^* = \{x_1x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A\}$$

These are operations on sets!

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

$A \cup B$: If A and B are regular lang, then $A \cup B$ is regular.

Proof:

What are we proving here?

- A. For any set A , if A is regular then so is $A \cup A$.
- B. For any sets A and B , if $A \cup B$ is regular, then so is A .
- C. For two DFAs $M1$ and $M2$, $M1 \cup M2$ is regular.
- ~~D~~ D. None of the above.
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1 , A_2 be any two regular languages over Σ .

WTS that $A_1 \cup A_2$ is regular.

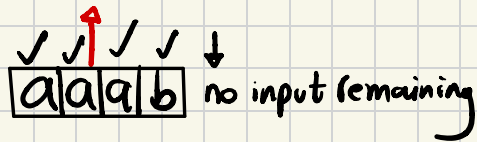
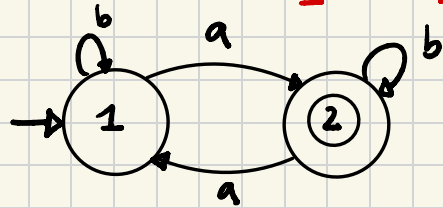
Goal: build a DFA that recognizes $A_1 \cup A_2$.

accepts

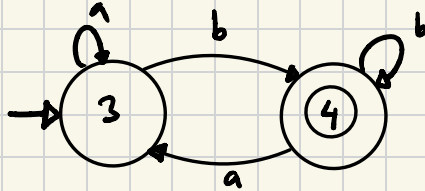
$A_1 = \{w \mid w \text{ has an odd number of } a\text{'s}\} = L(M_1)$

$A_2 = \{w \mid w \text{ ends with a } b\} = L(M_2)$

DFA For $A_1 : M_1$



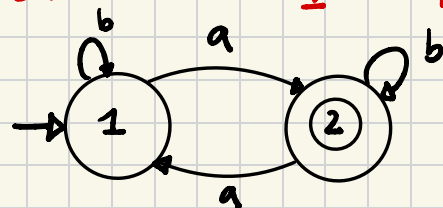
DFA For $A_2 : M_2$



$A = A_1 \cup A_2 \rightarrow \text{DFA } M$

$\hookrightarrow \{baaa, ab, aab, \dots\}$

DFA For $A_1 : M_1$



$$Q_{M_1} = \{1, 2\}$$

$$\Sigma = \{a, b\}$$

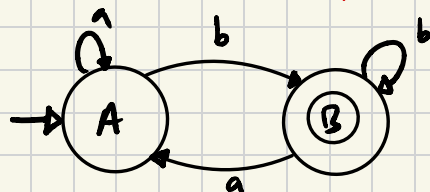
$$\delta: Q \times \Sigma \rightarrow Q$$

$$Q \begin{cases} \begin{array}{c|cc} & a & b \\ \hline 1 & 2 & 1 \\ 2 & 1 & 2 \end{array} \end{cases} \leftarrow \Sigma$$

$$q_0 = 1$$

$$F = \{2\}$$

DFA For $A_2 : M_2$



$$Q_{M_2} = \{3, 4\}$$

$$\Sigma = \{a, b\}$$

$$\delta: Q \times \Sigma \rightarrow Q$$

$$\begin{array}{c|cc} & a & b \\ \hline A & A & B \\ B & A & B \end{array}$$

$$\Rightarrow \delta(A, a) = A, \delta$$

$$\Rightarrow \delta(B, a) = A,$$

$$q_0 : A$$

$$F : \{B\}$$

DFA For $A : M$

$$A = A_1 \cup A_2$$

OR

$$Q = \begin{array}{c|cc} & A & B \\ \hline 1 & 1A & 1B \\ 2 & 2A & 2B \end{array}$$

$$Q = \{1A, 1B, 2A, 2B\}$$

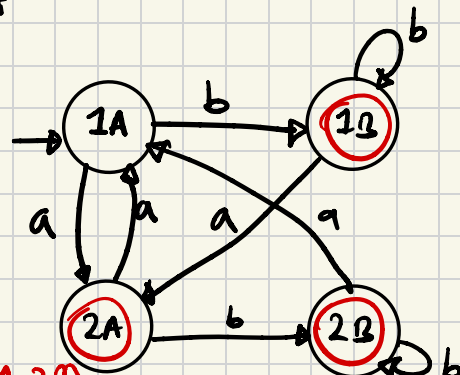
$$= |Q_{M_1}| \times |Q_{M_2}| = 2 \times 2 = 4$$

$$\Sigma = \{a, b\}$$

$$\delta: ((1, A), a) = (\delta_1(1, a), \delta_2(A, a)) = (2, A)$$

$$q_0 \rightarrow 1A$$

$$aaab$$



$$F = \{1B, 2A, 2B\}$$

$$\begin{array}{cc} 0 & 0 \rightarrow 0 \\ 0 & 1 \rightarrow 1 \\ 1 & 0 \rightarrow 1 \\ 1 & 1 \rightarrow 1 \end{array}$$

Union

Sipser Theorem 1.25 p. 45

Goal: build a DFA that recognizes $A1 \cup A2$.

Strategy: use DFAs that recognize each of $A1, A2$.



Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two regular languages over Σ .

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$

WTS that $A_1 \cup A_2$ is regular.

Define $M = (?, \Sigma, \delta, ?, ?)$

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two regular languages over Σ .

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$ and

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$. . .

Idea: run in parallel

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two regular languages over Σ .

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$. What should be the initial state of M ?

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$

What should be the initial state of M ?

- A. q_0
- B. q_1
- C. q_2
- D. (q_1, q_2)
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$:

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such th

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$

When r is a state in M_1 , s is a state in M_2 , and x is in Σ , then $\delta((r,s), x) =$

- A. (r,s)
- B. $(\delta(r,x), \delta(s,x))$
- C. $(\delta_1(r,x), s)$
- D. $(\delta_1(r,x), \delta_2(s,x))$
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$:

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such th

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$

The set of accepting states for M is

- A. $F_1 \times F_2$
- B. $\{ (r,s) \mid r \text{ is in } F_1 \text{ and } s \text{ is in } F_2 \}$
- C. $\{ (r,s) \mid r \text{ is in } F_1 \text{ or } s \text{ is in } F_2 \}$
- D. $F_1 \cup F_2$
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Proof: Let A_1, A_2 be any two regular languages over Σ .
Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and
 $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$.

WTS that $A_1 \cup A_2$ is regular. Define

$M = (Q_1 \times Q_2, \Sigma, \delta, (q_1, q_2), \{(r, s) \text{ in } Q_1 \times Q_2 \mid r \text{ in } F_1 \text{ or } s \text{ in } F_2\})$
with $\delta((r, s), x) = (\delta_1(r, x), \delta_2(s, x))$ for each $(r, s) \text{ in } Q_1 \times Q_2$
and $x \text{ in } \Sigma$.

Why does $L(M) = A_1 \cup A_2$?

Intersection

- *How would you prove that the class of regular languages is closed under intersection?*
- *Can you think of **more than one** proof strategy?*

$$\underline{A \cap B} = \{ x \mid x \text{ in } A \text{ and } x \text{ in } B \}$$

If A and B are regular, so is $A \cap B$

A regular $\rightarrow A^c$ regular

B regular $\rightarrow B^c$ regular

A^c regular, B^c regular $\rightarrow A^c \cup B^c$ regular

$$A^c \cup B^c = \bar{A} \cup \bar{B} = \overline{(A \cap B)}$$

$$\overline{A \cap B} \text{ regular} \Rightarrow \overline{\overline{A \cap B}} = A \cap B$$

$A \cap B$ is also regular.

If A and B is regular then,

So is $A \setminus B = A - B = A \cap \bar{B}$ is also regular.

Payoff

$\{ w \mid w \text{ contains neither the substrings } aba \text{ nor } baab \}$

Is this a regular set?

Payoff

$\{ w \mid w \text{ contains neither the substrings aba nor baab} \}$

Is this a regular set?

$A = \{ w \mid w \text{ contains aba as a substring} \}$

$B = \{ w \mid w \text{ contains baab as a substring} \}$

$$\bar{A} \cap \bar{B} = \overline{A \cup B}$$

General proof structure/strategy

Theorem: For any L over Σ , if L is regular then [the result of some operation on L] is also regular.

Proof:

Given name variables for sets, DFAs assumed to exist.

WTS state goal and outline plan.

Construction using objects previously defined + new tools working towards goal. Give formal definition and explain.

Correctness prove that construction works.

Conclusion recap what you've proved.

The regular operations

Sipser Def 1.23 p. 44

For A, B languages over same alphabet, define:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\} \quad \checkmark$$

$$A \circ B = \{xy \mid x \in A \text{ and } y \in B\}$$

$$A^* = \{x_1x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A\}$$



How can we prove that the concatenation of two regular languages is a regular language?